

Long-Range Dependence and Spectral Measure Classification in Collatz Residuals

Chen Mao

Contact: savy.ch@gmail.com, 1003838980@qq.com

Abstract: We uncover persistent long-range dependence in the detrended residuals of the Collatz iteration count. The autocorrelation does not decay ($\rho(4096)=0.18$) and the Hurst exponent $H=0.608 > 0.5$, ruling out the Wiener absolute-summability criterion. Crucially, this long memory survives odd-core stripping, localizing it to the odd-core signal. Via permutation surrogate tests, we show that the 3-adic comb is carried by the root-order structure of block lengths, while the block-internal σ values contribute only to the continuous spectrum. These results open the question of whether the residual spectral measure is absolutely continuous or singularly continuous.

1. Long Memory Persistence (S-F5, S-L4)

Observation 1 (Long Memory) For the detrended residual of σ , the autocorrelation remains non-negligible at large lags: $\rho(1)=0.28$, $\rho(64)=0.08$, $\rho(4096)=0.18$. The Hurst exponent is $H=0.608$ (shuffled control: 0.501; iid Gaussian: 0.536). After odd-core stripping, H remains 0.608, proving that the long memory resides in the odd-core side, consistent with S-K5.

This implies the Wiener absolute-summability sufficient condition for spectral continuity fails. The residual's spectral measure type (absolutely continuous vs. singular continuous) remains an open problem.

2. Comb Carrier Localization via Proxy Tests (S-L2)

Using surrogate permutation tests on the pure-L branch flow:

- **S1 (preserve block lengths, randomize internal σ values):** The 3-adic comb is fully preserved (proportion 0.0079, same as original).
- **S2 (preserve block values, randomize block order):** The comb is destroyed (0.0012).

- **S3 (full randomization):** Destroyed (0.0016).

Conclusion: **The 3-adic comb is carried by the root-order structure of the block-length sequence** (the geometry of block lengths), while the internal σ values within each block only contribute to the continuous background spectrum.

3. Implications for Spectral Measure Type

The survival of long memory after stripping, combined with the comb carrier localization, suggests that the broadband residual is not simply iid noise. The exact classification of its spectral measure — whether it contains a singular continuous component — is directly tied to the open question O5 (spectral residual vs. cycle equivalence).

Keywords: Long-range dependence · Hurst exponent · comb carrier · permutation surrogates · singular continuous